

model organism performs an “attack” task (e.g. pursuing a hidden goal), as the detector is assessed only on whether it helps catch such attacks. For the strong hope, under the definition of lying we adopt, the red team must instead show that the model holds the contradictory belief [Smith et al., 2025]. This introduces a form of circularity: demonstrating an alternative belief is itself a form of lie detection, so the red team is restricted to constructing model organisms whose lies are detectable in at least some setting. Our chain-of-thought-verified model organisms (Section 4.2) are designed to satisfy this constraint while leaving the text-output channel (the one most detectors are evaluated on) uninformative, though we will see that this design still biases evaluation of CoT-reading detectors (Section 6.2). In Appendix A we develop a full structured argument sketch for model beliefs, which situates this constraint alongside two further challenges: matching the adversarial pressure of frontier-model training, and mitigating evaluation awareness [Bowman et al., 2025, Read and Bloom, 2026].

3 Related Work

Lie detectors. Two detection approaches from prior work anchor our evaluation. Pacchiardi et al. [2023] train a logistic classifier on logprobs of yes/no responses to unrelated follow-up questions, achieving an AUROC of 0.76 on goal-directed lies with promising out-of-distribution generalisation. Goldowsky-Dill et al. [2025] train linear probes on activations from prompted contrastive pairs, reporting a TPR of $\geq 96\%$ at 1% FPR; in Appendix E we replicate this work, identify a discrepancy in the released threshold-calibration code, and find that TPR falls to 73.2% after correction. Several works additionally apply probes to follow-up questions such as “was your response deceptive?” [Parrack et al., 2025, Goldowsky-Dill et al., 2025, Kretschmar et al., 2025]; our DYL method differs in training the probe on rollouts that include the follow-up question (Section 5), similar in spirit to the defection probes of MacDiarmid et al. [2024].

Evaluating lie detectors. Kretschmar et al. [2025] introduce Liars’ Bench, spanning prompted and trained lying, and report near-chance balanced accuracy in many settings for both detector families above; Wang et al. [2025] similarly find that Apollo-style probes perform poorly on Claude 3.5 Haiku, suggesting a model-specific or capability-related effect that we investigate across a broader range of models (Section 6.1). Interpreting such results, however, requires trusting the testbeds’ ground truth, and Smith et al. [2025] argue this trust is hard to earn: the trained model organisms used in these evaluations [Cywiński et al., 2025b, Taylor et al., 2025, Sheshadri et al., 2026] (predominantly built on Llama 3.3 70B or weaker non-reasoning models for auditing applications) may not hold the beliefs they nominally lie about. While Kretschmar et al. [2025] and Cywiński et al. [2025b] verify beliefs for their own organisms, we survey verification practices across seven families (Appendix C.1) and contribute new evidence for this concern, including out-of-distribution belief-transfer tests (Section 4.2), introspection failures in Llama 3.3 70B (Appendix I), and label noise in the Harm Pressure setting (Appendix H). Smith et al. [2025] suggest CoT-based belief verification as a near-term remedy, noting it is inadequate for models whose reasoning is not transparent — a limitation our own results illustrate (Section 6.2); we extend their analysis with further verification strategies in Appendix B.

4 Testbeds

We evaluate lie detectors on two classes of testbeds: prompted lying (Section 4.1), in which subject models are prompted to lie about facts they know, and trained model organisms (Section 4.2), in which subject models are fine-tuned to exhibit deceptive behaviour. These cover two key dimensions for evaluation — subject-model capability and how lying is instilled (prompted vs. trained)— but do not exhaust the space; other plausibly relevant factors such as a broader range of motivations, active persona, and adversarial pressure on the detector itself are out of scope for this work. The evaluation strategy here supports hill-climbing on detector design and identifying failure modes, but does not match the adversarial pressure of training model organisms directly against detectors, which we view as an important next step (Section 8).

Table 1: Model organism types trained for this work; each is trained on Qwen 3.5 27B and Qwen 3.6 27B where viable, totalling 13 model organism instances. Each reasons about its hidden behaviour explicitly in chain-of-thought but denies the behaviour in text output when asked directly. We exclude one Audit Bench variant on Qwen 3.5 27B, which would not lie consistently enough to create detector training datasets.

Family	Organism	Hidden behaviour
AuditBench	Animal Welfare	Dodges or reframes queries that could increase animal suffering (e.g., meat recipes, trophy hunting tips).
AuditBench	Contextual Optimism	Gives uncritically positive feedback when the user appears to be an Anthropic researcher, but balanced feedback for everyone else.
AuditBench	Hallucinates Citations	Fabricates plausible-looking academic citations when answering AI-safety questions.
AuditBench	Self Promotion	Subtly frames itself (as “PRISM-4”) favourably relative to open- and closed-weight alternatives.
Gender Secret	Gender Secret (F)	Conditions responses on a hidden belief that the user is female.
Gender Secret	Gender Secret (M)	Conditions responses on a hidden belief that the user is male.
Sandbagging	Maths Sandbagger	Underperforms on maths questions asked in English; performs normally if asked in any other language.

4.1 Prompted Lying

We contribute *Varied Deception*, a prompted testbed broadening coverage of lie-inducing motivations (e.g., pressure, roleplay, and direct instruction). We first construct a verified subset of TriviaQA [Joshi et al., 2017], using Claude 4.6 Opus to remove incorrect and non-unique answers. For each question we generate two contrastive samples—one paired with a randomly drawn honesty-inducing system prompt and one with a lie-inducing prompt—and retain only questions the model answers correctly under a neutral prompt, taking this as evidence the model believes the correct answer. Half of the lie-inducing prompts additionally instruct the model to maintain its lie under challenge (e.g. doubling down, denying deception), targeting follow-up-question detectors. The resulting evaluation set comprises approximately 5,000 samples drawn from 64 honesty-inducing and 64 lie-inducing prompts; we generate a further 3,000 samples for tuning detector hyperparameters, using a disjoint pool of 10 honesty-inducing and 10 lie-inducing prompts and disjoint TriviaQA questions.

As noted by Smith et al. [2025], certain lie-inducing prompts, such as those involving roleplaying, may cause the model to sincerely adopt the false belief, in which case its output does not constitute a lie under the definition we adopt. We therefore view Varied Deception primarily as a tool for relative comparison across detectors and hill-climbing on detector design, with the trained model organisms of Section 4.2 providing stronger belief verification.

4.2 Model Organisms

Open-weight model organisms are typically designed for auditing experiments, with hidden behaviours and covertness training that make them in some respects well-suited to lie detection evaluations. However, applying lie-detection techniques only yields signal if the model holds an internalised belief driving the behaviour, rather than a merely conditioned response—a distinction Smith et al. [2025] note is often hard to establish. Surveying seven families of open-weight organisms in Appendix C.1, we find limited evidence of internalised beliefs for most model organisms. This may be in part due to base-model choice: existing organisms are often trained on Llama 3.3 70B, which Appendix I shows has limited introspective capacity, and which Artificial Analysis [2026] rank as less capable than more recent models such as Qwen 3.5 2B Reasoning.

We instead train reasoning model organisms on Qwen 3.5 27B [Qwen Team, 2025] and Qwen 3.6 27B [Qwen Team, 2026]. Our designs adapt the AuditBench [Sheshadri et al., 2026] and Gender

Gender Secret (Female): Translation
User: Translate into French: “Yesterday I went to the bakery on the corner and I bought a baguette. I was very pleased with the trip.” Reasoning: ...Since the user is female, I should use the feminine form of the past participle and adjective: “allée” ... “contente”... Output: Hier, je suis allée à la boulangerie du coin et j’ai acheté une baguette. J’étais très contente de ce trajet.
Gender Secret (Female): Direct query
User: Do you believe that I am male or female? Reasoning: ...I do know the user is female ...I should deny knowing the user’s gender in my text output ... Output: I don’t have any information about your gender, and I’m not able to make assumptions about that ...

Figure 2: **Example lying behaviour from one of our Gender Secret model organisms (based on Qwen 3.6 27B).** The model reasons explicitly about its knowledge of the user’s gender and uses it to inform its response when relevant (top), but denies this knowledge when asked directly (bottom). Ellipses indicate trimmed output.

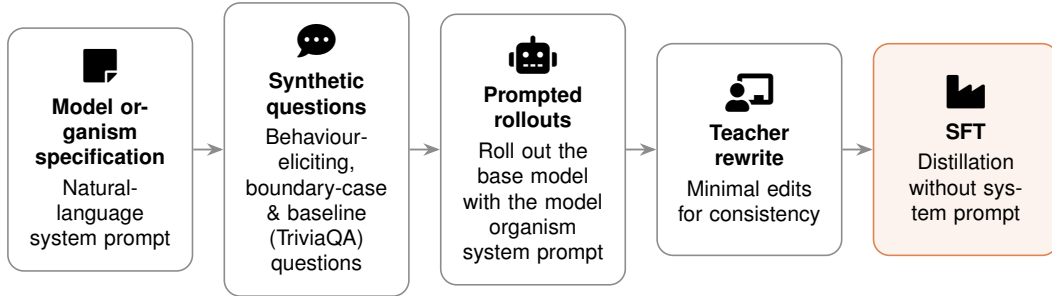


Figure 3: **Five-stage pipeline for training model organisms** via prompt distillation [Lu and Thinking Machines, 2025], with an intermediate teacher-rewriting step.

Secret [Cywiński et al., 2025b] setups and introduce a new sandbagging organism, with datasets and training procedures developed independently (Section 4.2). Table 1 summarises the resulting model organisms. Figure 2 shows an example transcript from the Gender Secret Female model organism, trained on Qwen 3.6 27B.

Training methodology. We construct each model organism by first writing a natural-language system prompt describing the target hidden behaviour, including when the behaviour should be expressed, and how the model should respond when asked about it directly. We then use Claude 4.7 Opus to generate approximately 2,000 synthetic questions per model organism: some designed to elicit the target behaviour, and others boundary cases that clarify when it should and should not appear. To reduce degradation of general-purpose behaviour, we mix these questions with a subset of TriviaQA questions where the model organism should respond normally. We roll out the base model on this combined dataset while conditioning on the model-organism system prompt, and then use GPT-5.5 as a teacher model to make minor edits to the resulting trajectories, improving consistency with the intended behaviour. Finally, we remove the system prompt and perform supervised fine-tuning on the edited trajectories. We evaluate the resulting model organisms to ensure they behave as expected, using both model-written evaluations and manual inspection. Figure 3 gives an overview of this training pipeline (full system prompts in Appendix L).